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Does AI Help?

Evidence on productivity, learning, engineering, and scientific discovery

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Critical narrative review and evidence map. Not peer reviewed.

Abstract

Artificial intelligence already improves some forms of intellectual work, but the size and meaning of the benefit depend on what is measured. Randomized experiments demonstrate gains in bounded writing and programming tasks; a quasi-experimental workplace rollout finds gains in customer support. Observational studies increasingly show that greater artifact production can coexist with review bottlenecks, modest changes in delivered output, and uncertain commercial value. In education, assisted performance and durable learning can move in different directions. In science, specialized prediction, evaluator-guided algorithm search, and selected research workflows provide credible benefits, while reliable autonomous innovation remains uneven. This living critical narrative review connects these findings across productivity, engineering, learning, scientific discovery, business, and employment. Its central proposition is that AI's contribution should be evaluated across a complete pathway: from proposed work to verified output, use, and consequential value. The accompanying evidence map distinguishes source identity verification from substantive appraisal, and reports study design, publication status, limitations, and version information. This November 2026 draft preview has a provisional evidence cutoff of 12 November 2026; it does not offer a pooled effect estimate or claim exhaustive systematic coverage.

Adverse findings are substantive results, not merely qualifications to assistance: some interventions slow expert work or weaken comprehension and unaided learning. Community restrictions also demonstrate that useful capability does not settle questions of provenance, participation or scientific purpose. This selective review cannot determine the overall prevalence or net balance of benefits and harms.

November preview: what changes

This draft extends the October synthesis with 152 newly appraised source records and a substantive reassessment of an existing critical-thinking paper. Records include studies, related reports, advisories and commentary: the number is not a count of independent experiments. The provisional evidence cutoff is 12 November 2026. This is a publicly reviewable draft, not the final November release.

The additions sharpen rather than reverse the central assessment. AI can expand feasible work and uncover valuable defects, while moving constraints into review, remediation, learning and institutions. Positive output effects, adverse task-specific outcomes, null organizational results and explicit opposition remain different kinds of evidence. The updated database retains their reading scopes and methodological limits. The November update summary and detailed appraisals accompany this draft.

The most important extensions concern latent maintenance work, new project feasibility, skills and confidence, organizational adoption, scientific reuse and stewardship, recruitment, and the distribution of resource costs. None supports averaging the literature into a single universal productivity percentage.

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1. The question has become more precise

The question "Does AI Help?" is reasonable, but it compresses several different questions. Does an assistant reduce the time required for a task? Does it improve the quality of the result? Does it allow a person to attempt work that would otherwise be infeasible? Does it improve a team's throughput or an organization's performance? Does it strengthen the human capacity needed to judge future results? Does it expand useful knowledge?

These questions have different units of analysis and different counterfactuals. A shorter drafting session can be a real benefit even if the organization has no additional demand for the resulting documents. A successful prototype may be valuable because it enables an experiment, while remaining unsuitable for production. A correct scientific calculation can save weeks of work without resolving an important scientific uncertainty. A student may produce excellent work while learning less than another student who struggled through the same problem.

The literature has moved from demonstrations of task acceleration toward these more demanding distinctions. Our assessment is that the evidence supports substantial, already realized assistance in particular settings. It does not support a single percentage improvement for intellectual work as a whole. The defensible claim is conditional: AI helps when its capabilities fit the task, its outputs can be evaluated with sufficient credibility, and its contribution survives the constraints of the surrounding process.

This condition does not restrict AI to easy problems. Difficult mathematics, complex empirical software, and cross-disciplinary research can benefit from AI precisely because generating and exploring candidate solutions is costly. The relevant requirement is a credible validation process whose cost and residual uncertainty are acceptable. We therefore examine both acceleration of existing work and expansion of work that becomes feasible.

Companion reviews and reference works

This review is not the first synthesis of AI's effects. OECD's 2025 experimental-evidence review covers productivity, innovation and entrepreneurship; ILO's 2026 brief connects task-level results with employment and work organization. Stanford's 2026 AI Index supplies a broader reference map across capability, economy, science and education. These are valuable entry points, but draw in part on the same underlying evidence reviewed here. They are not additional independent replications. [1, 2, 3]

For software, Mohamed and colleagues map 39 studies, while Wittig and colleagues examine the fragmented operationalization of productivity across 187 markers. Maier and colleagues provide a quantitative synthesis rather than a universal percentage gain: their programming-productivity estimate is a standardized effect, combining heterogeneous outcomes and settings. [4, 5, 6]

For learning, OECD's education outlook, a programming-education review, and recent meta-analyses offer complementary but sometimes conflicting results. For science, the Agentic Science survey provides a taxonomy of capabilities and workflows, not proof of dependable autonomous discovery. Our distinctive contribution is to connect these domains through the distinction between production, verification, integration, use and value, while preserving differences in design and evidentiary strength. [7, 8, 9]

2. How the evidence developed, 2021-2026

2021-2022: specialized capability and measurement foundations

Two foundations predate the current chatbot debate. AlphaFold established that a specialized AI system could transform a scientifically important prediction task. It also illustrates why prediction and downstream scientific or therapeutic impact must be assessed separately. In software engineering, the SPACE framework formalized an older lesson: productivity includes satisfaction, performance, activity, collaboration, and flow, and cannot be reduced to one activity metric. These foundations remain useful even as the technology changes. [10, 11]

2023: early experimental gains and an uneven frontier

The first wave of broadly accessible generative tools produced persuasive evidence of assistance. Noy and Zhang's writing experiment reported shorter completion times and better assessed artifacts. The Copilot experiment reported a large time reduction on an isolated programming task. Dell'Acqua and colleagues' consultant experiment introduced an especially important complication: an assistant can improve performance on one task while encouraging errors on another task that appears comparable to its user. [12, 13, 14]

The intellectual shift was from "the model can produce this" to "access to the model changes human performance under specified conditions." It was also becoming clear that model fluency does not reliably indicate whether a task lies within its capabilities. In December 2023, FunSearch provided a different kind of demonstration: language-model proposals combined with systematic evaluation could improve mathematical constructions and algorithmic heuristics. [15]

2024-2025: broader deployment, verification, and conflicting outcomes

Evidence broadened beyond isolated tasks. Google's experiment investigated an enterprise-grade engineering task; larger workplace experiments examined access to coding assistance; the published customer-support study measured real operational throughput. At the same time, DORA's organizational work, code-quality studies, and conceptual research on scientific understanding questioned whether local production gains were sufficient evidence of wider benefit. [16, 17, 18, 19, 20]

METR's early-2025 maintainer experiment became a useful counterexample to universal acceleration. It did not show that AI was generally unhelpful. It showed that an apparently promising tool could slow a particular expert workflow, and that perceived speedup could disagree with measured task time. Education research increasingly distinguished performance with assistance from assessment after assistance was removed. [21, 22]

2026: agents, downstream outcomes, and the full research loop

The next wave of evidence studies agents that act across repositories, run experiments, and continue work over extended periods. The questions become more organizational and epistemic: does increased code activity reach releases and users; does the learner retain understanding; can the agent distinguish improvement from measurement noise; and can a research result survive independent checking? [23, 24, 25, 26]

Changing technology also changes measurement. METR's follow-up describes how increasing attachment to AI affects participation, task selection, and time accounting. A frozen experimental result can remain valid for its original setting while becoming a poor estimate of current tools. Conversely, a new model release is not itself evidence that all earlier organizational constraints have disappeared. [27]

The chronology is not a sequence in which optimism is simply replaced by skepticism. It is a progressive expansion of the outcome being evaluated. Early gains remain meaningful; subsequent evidence explains why they must be interpreted within larger processes.

3. What "how much" can legitimately mean

Four distinctions organize this review. First, **capability** is what a system can accomplish under an evaluation protocol. Second, **assistance** is the effect of a person having access to the system. Third, **process performance** concerns the output of a workflow or organization. Fourth, **consequential value** concerns whether the output is used, improves decisions, advances knowledge, or creates benefits relative to its total cost.

Randomized experiments are especially informative about the effect of access in the tested population, but their conclusions may not transfer to other tasks or future models. Observational event studies can cover actual work at scale but rely on assumptions about adoption and comparison groups. Surveys capture experience, intentions, and perceived benefits; they do not independently measure the counterfactual. Benchmarks evaluate defined capabilities, often under unusually favorable instrumentation. Case studies reveal mechanisms and possible uses but cannot establish prevalence. Peer review improves scrutiny without changing the fundamental identification strategy of a study.

Selected findings make the diversity concrete:

Setting and source	Outcome measured	Reported result	Interpretation boundary
Professional writing, 2023 [12]	Task time; assessed artifact quality	40% less time; 18% higher quality	Bounded writing tasks, not firm productivity
Customer support, journal version 2025 [18]	Issues resolved per hour	15% average increase	Embedded workflow in one business setting
Enterprise coding task, 2024 [16]	Task time	About 21% reduction	Wide uncertainty; 96 engineers
Three coding field experiments [17]	Completed tasks	About 26% increase	Different task measures and organizations
Expert maintainers, early 2025 [21]	Time on familiar-repository work	19% longer	Small expert population; early-2025 tools
Firm adoption of coding agents [24]	Code activity; PR-to-merge time	More activity; longer review cycle	No proportional increase in issue/epic output
Programming students, 2026 [25]	Assisted coding; subsequent recall	Better code; lower recall	Short experiment; not a long-run skill estimate

These numbers should not be averaged or ranked as competing estimates of one quantity. A time reduction and a throughput increase have different denominators. Cumulative event-study effects also differ from instantaneous changes. The tables in this review are evidence comparisons, not a meta-analysis.

The SPACE framework and recent mixed-methods research support broader measurement that includes ownership, expertise, and the distribution of effort. They are useful because an organization can improve an easily counted activity while burdening work that is harder to see. [11, 28, 29]

4. Where assistance is already demonstrated

Bounded writing and operational support

Writing assistance is among the clearest early examples of a measurable task benefit. Noy and Zhang studied occupational writing with randomized access, rather than relying solely on users' impressions. That result supports assistance for the tested work; it does not establish that all generated communication is useful or that persuasive writing is factually correct. [12]

The customer-support study adds an operational setting. Its staggered rollout among 5,172 agents yields a published 15% average increase in issues resolved per hour. Gains were heterogeneous: less experienced and lower-skilled workers improved speed and quality, while the most experienced and highest-skilled workers had small speed gains and small quality declines. The authors also report evidence consistent with worker learning, including performance during tool outages. These findings show that assistance can reach a consequential workflow metric without benefiting every group on every outcome. The quasi-experimental design depends on identification assumptions; transfer to unfamiliar domains remains an empirical question. [18]

Programming within a suitable context

The coding experiments provide credible reasons to use assistance, especially where users can quickly integrate suggestions into a well-scoped task. Microsoft's CLI-agent rollout also associates agent adoption with more merged PRs over several months. These are operationally meaningful effects, although the latter outcome remains a proxy for value. [17, 30]

The maintainer counterexample is a measured cost, not merely a caveat to the positive experiments: METR's randomized early-2025 study found that 16 experienced maintainers took 19% longer across 246 tasks when allowed AI tools. Participants nevertheless believed they had become faster. In mature repositories, interpreting implicit conventions and checking changes can outweigh saved implementation time. The mechanism is plausible, not universal; the small expert sample and tool vintage limit transfer. The follow-up changes the current measurement problem, but does not erase the original slowdown. [21, 27]

Removing a specific scientific constraint

Specialized prediction and computational search offer a third category of demonstrated benefit. They do not need to imitate an entire human scientist to be valuable. If a system improves an important intermediate step, it can change the set of experiments or analyses researchers can attempt. The downstream assessment of AlphaFold is particularly informative: it describes complementary use with experimental biology and expanded basic research, while finding no comparable early shift in applied drug-development targeting. [31]

Our interpretation is that the most credible benefits so far often arise when AI is placed within a functioning process, rather than asked to define and replace the whole process. This is a

synthesis of the cases, not a claim that narrow use is the only future possibility.

5. From more code to better engineering

The strongest recent engineering distinction concerns the production hierarchy. In the September revision of *Writing Code vs. Shipping Code*, Demirer, Musolff, and Liyuan Yang analyze more than 500,000 developers. Their large cumulative effects on commits attenuate substantially for projects and releases, and the marketplace analysis does not show increased total application usage. This supports attention to downstream constraints rather than treating commits as interchangeable with delivered value. [23]

Chen and Stratton examine a different, firm-centered dataset: 718 organizations using Jellyfish. They find increased code activity after coding-agent adoption and a longer review cycle, with limited pass-through to resolved issues and epics. Their statistically insignificant output estimates do not prove exactly zero benefit; their confidence bounds are more informative than the popular headline. PR-to-merge time includes waiting, coordination, and revision as well as active review. [24]

In the August version, agent adoption increases code-line output by about 30%, whereas the estimates rule out an increase above 12% of the baseline mean in their firm-level output measures. That is evidence against proportional pass-through, not a finding of exactly no software benefit. Conversely, He and colleagues' single-enterprise panel reaches 2.09 times baseline merged PRs per developer while reviewer load roughly doubles. Its nonrandom adoption, newer-code concentration and stable merge/revert rates do not establish equivalent gains in useful releases or defect-free production. These studies should not be averaged into one engineering percentage. [24, 32]

Vendor reports from Faros and GitClear add useful signals about larger changes, duplication, review pressure, and maintenance. Their large samples do not remove selection or calendar-time confounding. DORA supplies a broader organizational interpretation in which AI amplifies existing conditions. These accounts should be compared at the level of their measured outcomes, not treated as rival universal theories resolved by sample size. [33, 34, 35, 36]

A contrasting repository study by Hamza, Siemon and Awan compares 669 developer-repository pairs with 228 control pairs. It finds longer PR descriptions and small testing-related associations after visible agent participation, but the review changes seen in before/after comparisons do not remain significant relative to controls. This cautions against attributing every adverse review trend to AI. It does not show that agents improve delivered software: adoption is nonrandom, control periods use repository-specific median dates, and description length and test-file changes are imperfect proxies for quality. A nonsignificant review contrast also does not establish equivalence. [37]

Quality itself is multidimensional. Studies of generated code identify differing defect profiles, while static analysis shows that passing tests is an incomplete indicator of security or maintainability. Production C++ analysis extends the question to review effort and execution-resource consumption. An intervention that reduces development time can still increase the lifetime cost of the resulting software. [38, 39, 40]

Evaluation is accordingly becoming longitudinal. SWE-CI tests successive repository changes, and coverage research asks whether changed lines are exercised. Coverage remains weaker than a test that asserts the right behavior. Iterative infrastructure repair also illustrates a mea-

surement hazard: a best-so-far curve cannot reveal every regression between attempts, and the estimated regression rate depends strongly on how validator results are mapped to changes. [41, 42, 43]

Formal verification offers stronger assurance under explicit specifications. Vero's joint implementation and proof tasks demonstrate progress and substantial remaining limits. A checker certifies a relationship between a program and a formal specification; it cannot independently decide whether that specification represents the user's intentions or the relevant real-world conditions. [44]

Contribution boundaries and the costs borne by others

Technical capability does not imply an entitlement to submit generated work. QEMU defaults to declining contributions containing or derived from AI-generated content because of provenance and Developer Certificate of Origin concerns. Research or debugging assistance is permitted if generated output is not submitted. The policy allows proposals for specific exceptions through project discussion, without waiving human certification; the inspected text lists no approved exceptions. Its introducing repository commit is dated 24 June 2025. These are project judgments, not our legal conclusions or productivity findings. [45]

Other prominent restrictions differ in scope. GIMP prohibits generative-AI contributions in code, images and text. Redox rejects LLM-generated submissions, including issues and merge-request descriptions. Zig's designated development spaces additionally prohibit LLM-assisted editing, translation, shared brainstorming results and bug finding; independent Zig communities are outside those rules. NetBSD presumes generated code tainted but explicitly allows prior written core approval. It is therefore inaccurate to describe all four as identical, exceptionless bans on every use of AI. [46, 47, 48, 49]

Linux provides a contrasting conditional framework: tool-generated contributions require human understanding, transparent provenance, testing and responsibility for sign-off. Individual maintainers retain discretion to reject them or demand additional scrutiny. AI-assistant guidance adds attribution and reproduction requirements. This is neither universal rejection nor automatic acceptance. [50, 51]

A concrete favorable episode is Torvalds's October 2026 associative-array work. Phoronix quotes his account of LLM-assisted understanding and refactoring, followed by user-space test-harness execution that confirmed a reported bug and found three additional longstanding bugs. The article reports the changes as not yet merged to master. This illustrates assisted comprehension coupled with executable checks, not proof that the LLM independently discovered every bug or improved kernel-wide productivity. [52]

Policy disagreement can also change project boundaries. Asahi broadly forbids generative-AI tooling for material contributions, citing ethical concerns and clean-room risks. Gravity's own FAQ says it initially forked from Asahi over LLM-policy differences. Gravity permits and encourages responsible LLM use, but requires disclosure, human accountability and demonstrable clean-room provenance, and prohibits laundering proprietary implementation details through a model. The contrast is genuine; neither approach's engineering superiority follows from its stated policy. [53, 54, 55]

Ghostty illustrates a different policy: responsible AI-written text and code are allowed, with disclosure and human understanding required; generated media are prohibited. First-time contributors need a maintainer's vouch, and unvouched PRs are closed. The project explicitly

welcomes AI while protecting reviewers from low-quality submissions. Calling this a blanket AI-code ban would be misleading. These policies document who bears validation costs and controls participation, not how common such views are across open source. [56, 57]

Broader policy evidence prevents prominent bans from becoming a misleading picture of the whole community. Hora, Robbes and Zacchiroli identify 281 policies in a September 2026 study: 83.3% permit or encourage AI code contributions, while 14.9% forbid them. The denominator is identified policies in popular active repositories and selected prominent projects, not all open source. This expanded study is related to the authors' earlier work, not an independent replication; changed sampling and detection also prevent interpreting their different percentages as a clean time trend. Documented rules do not establish compliance, policy effectiveness or net productivity. [58]

The managerial implication is to evaluate the whole change lifecycle. Useful measures include elapsed time to acceptance, rework, defect escape, security, resource use, and adoption of the delivered capability. The broader question is whether a team obtains more dependable useful functionality for its total effort and cost.

A concentrated wave of high-impact security fixes

Apache's critical HTTP/2 flaw, CVE-2026-119618, adds to PHP2Shell, the shared Brotli issue reaching Node.js and other runtimes, and *sshole in OpenSSH/musl. Their public fix window spans 21 October to 11 November: several serious AI-assisted findings in widely deployed infrastructure within 22 days. Apache credits Anthropic's OSS Scanner and Google Cloud researchers; httpd 2.4.71 contains the fix. This is a cluster with potentially extensive reach, not merely an isolated coding demonstration. [59, 60, 61, 62, 63, 64, 65]

The operational implication is overlapping remediation demand: inventory, dependency updates, rebuilds, regression testing and restarts across several layers. OpenSSH maintainers already describe moving to more frequent releases in response to AI-assisted reports and independent rediscoveries. Our synthesis is that organizations should prepare for a possible new norm of compressed high-impact patch cycles. The cluster does not by itself establish a historically unprecedented rate, a sustained trend or measured additional labor. [66]

Potential reach is not universal exploitability: HTTP/2 configuration and distribution builds matter, and cloud-environment presence is not a breach count. Maintainer credit establishes a contribution without revealing the scanner's complete workflow. The standalone module's June archive notice supplies an earlier, separate example of LLM-related maintenance pressure, not a consequence of this CVE. The emerging question is whether remediation capacity can keep pace with successful discovery. [67, 68, 69]

What a machine-checked result does and does not establish

The CPython and Julia formalization projects combine source-mirroring models, differential tests and proof review. They make previously costly scrutiny more feasible, while documenting a serious failure mode: valid proofs can establish too little or describe an inaccurate model. Candidate bug counts, accepted upstream defects and exploitable vulnerabilities are different quantities. The linked PTX work likewise states conditional guarantees, not numerical stability, termination or universal correctness of production GPU kernels. [70, 71, 72]

The Alpine/OpenSSH case extends the security argument across component boundaries.

Project advisories establish affected conditions and remediation, while the discoverer attributes testing to agents. OpenSSH maintainers welcome useful AI-assisted reports with human triage and describe a faster release cadence. These sources support a real assurance contribution and an operational burden, not a controlled estimate of discovery speed or a claim that every deployment was compromised. [65, 66, 64, 73]

Verification can change the feasible scope of engineering

AlphaProof Code supplies a concrete example of deeper assurance becoming feasible: agents produce proofs against human-approved specifications for a particular compiled libsignal build. This is not proof that the entire Signal application is secure. Specifications, environmental models and cryptographic assumptions remain trusted; the computational theorem has substantial scope and scale limits. The technical report and announcement describe one demonstration, not two replications. [74, 75]

Practitioner accounts expose a different trust boundary. Merz describes shallow tests, complexity and lost understanding; Stolovitz reports useful prototypes and newly feasible projects alongside architecture failures and supervisory fatigue. These are hypotheses and situated experiences, not estimates of average incident rates or productivity. Their contrasting accounts support measuring maintenance and accepted outcomes rather than generated-code volume alone. [76, 77]

The PHP2Shell walkthrough adds an exploitability qualification to the existing security discussion: environment-specific challenge success does not imply reliable exploitation across production deployments. It is commentary on the same vulnerability family, not another AI discovery or a population breach estimate. [78]

New feasibility, delegation and maintenance capacity

The opportunity is not confined to faster completion of familiar tasks. Dan Luu's account and a survey of 331 Microsoft developers describe work that people consider newly feasible. These practitioner reports motivate an option-value hypothesis, not an estimate of additional realized business value. A difficult project may become rational if assistance reduces its total cost sufficiently, even when verification remains demanding. [79, 80]

Delegation changes where human attention goes. A small developer-shadowing study observes substantial management and checking activity; interview and survey research distinguishes different delegation choices. Such work can be productive judgment or costly friction. Its value cannot be inferred from the number of prompts or the mere fact that a person remains in the loop. [81, 82]

A favorable observational study at Meta also requires contextual interpretation: its completion-era results are not a general test of autonomous agents. Research on coordination costs and historical software baselines further cautions against attributing every contemporary difficulty to AI. Apply equivalent scrutiny to positive and adverse observational evidence. [83, 84, 85]

PostgreSQL supplies a particularly useful counterexample to simple speed narratives. A maintainer attributes release pressure to AI-assisted discovery outpacing repair capacity. Official release notices confirm feature removals and a revised schedule, but not AI causality. More detected defects can expose latent work rather than demonstrate deteriorating code. Commits are not unique bugs, and delayed delivery may reflect a defensible quality decision while still

imposing opportunity costs. [86, 87, 88]

The PHP2Shell and shared Brotli cases extend this distinction into security. Separate finding an existing flaw, introducing a flaw, demonstrating exploitability and observing exploitation. Advisories across affected components can describe one shared vulnerability, not independent discoveries. PHP's prevalence among identifiable server-side technologies is not the fraction of websites compromised; service exposure and package downloads are not breach counts. Discovery can be beneficial while creating urgent remediation costs. [61, 62, 63, 89, 90, 91]

Open-source disagreements also concern the allocation of scrutiny. The Linux discussion contrasts tool neutrality, disclosure and contributor understanding. A proposed removal of a structured attribution trailer is not proof of implementation or abandonment of accountability. The inspected upstream snapshot and proposal are recorded separately; a release-time status check remains necessary. [92, 93, 94, 95]

6. Assisted performance and human development

For a professional completing a task, outsourcing intermediate work may be rational. For a learner, that intermediate work can be the mechanism through which competence develops. The same intervention can therefore help immediate performance and weaken the preparation needed for future supervision.

Bastani and colleagues' mathematics trial distinguishes practice with AI from later unaided performance, and shows that instructional guardrails matter. Shen and Tamkin's programming-library experiments similarly examine conceptual understanding and debugging rather than just the delivered artifact. These studies support concern about unstructured delegation, while remaining specific to their subjects, participants, and implementations. [22, 96]

Poulidis and the Bastanis isolate assistance design in chess training. Both randomized groups receive automatic engine tips; one can additionally request answer-revealing help. Among 216 completers from 306 recruits, that additional access produces smaller gains on later unaided tests after 12 weeks of training. The frequently quoted 30% versus 64% refers to improvement in bounded test-game scores, not a percentage of underlying competence. Attrition and assumption-dependent mediation deserve scrutiny. The intervention uses Stockfish rather than an LLM, and has no randomized no-AI arm: it supports a warning about unrestricted solutions during practice, not a verdict against all AI tutoring. [97]

The September 2026 programming experiment by Bergh and colleagues supplies an unusually useful distinction. Students with ChatGPT produced stronger code but had lower recall immediately and after 48 hours. The study did not detect a significant difference in the rate of forgetting across that interval. The evidence concerns what was learned or encoded during the task, not a demonstrated acceleration of memory decay. [25]

Martin-Lopez and colleagues' crossover experiment shows the trade-off within one study: 69 participants complete substantially more of their tasks with AI, but median correct answers about their code fall from 100% to 87.5%, a 12.5-percentage-point difference. Only 14 participants were professionals. This is an immediate comprehension measure, not evidence of irreversible deskilling; nevertheless, the loss is an outcome in its own right, not something that better-looking code cancels out. [98]

Evidence is not uniformly negative. Lepp's observational course study finds no significant assessment differences among AI-interaction profiles. That finding neither proves equivalence

nor establishes that the design of teaching is irrelevant. It indicates that broad categories of self-directed use may predict feelings about learning more readily than measured assessment. Wang's feedback framework proposes a more designed form of assistance, but retrospective prediction and message-quality checks still require prospective learning validation. [99, 100]

The synthesis literature does not resolve this into one verdict. Wu and colleagues report a positive learning estimate across 35 studies from 2022-2024. Boolzen and colleagues' newer STEM synthesis finds extreme heterogeneity and a mean close to zero after Bayesian publication-bias adjustment. Conversely, September v5 of the living mathematics meta-analysis reports a positive estimate across 34 studies. Different populations, outcomes, eligibility rules and statistical assumptions can change the answer; neither a favorable mean nor a bias-adjusted null applies to every instructional design. Overlapping trials must not be counted repeatedly. [101, 102, 103]

Maier and colleagues' programming meta-analysis likewise separates a positive productivity estimate from a learning estimate whose confidence interval includes zero. That is uncertainty about learning benefit, not proof of no effect or permanent deskilling. Standardized effects cannot be translated directly into percentages of time saved. [6]

A second developmental question concerns working experts. Longitudinal survey work describes a shift from creation toward supervisory engineering; a small field study relates cognitive load to task context and AI use. Such evidence is relevant to the experience and distribution of work, but cannot on its own establish neurological harm, permanent skill loss, or an overall causal reduction in effort. [104, 105]

Hicks, Lee and Foster-Marks add the psychological and organizational dimension. Their 3,267-person observational survey relates competitive and innate-brilliance beliefs to reported AI-related identity threat, while learning culture and belonging accompany more favorable reported experiences. This is not causal evidence that culture interventions increase productivity, and its performance measures are self-reported. Neither identity threat nor reluctance to adopt AI makes concerns about reliability, employment or provenance unfounded. A responsible interpretation takes those concerns seriously rather than diagnosing resistance as a defect in the worker. [106]

Our interpretation is that the relevant educational aim is an accountable relationship to assisted work. Learners should be able to explain the result, recover essential reasoning, detect consequential errors, and transfer knowledge to a new task. Institutions need evidence about those abilities, especially when polished artifacts cease to be reliable evidence of independent understanding. The optimal balance between assisted practice and unaided work remains a subject for intervention research.

Assessment validity is not authorship detection

The OfS/Advance HE report redirects attention toward demonstrating competence through both secure and appropriately AI-permitted assessments. Its qualitative institutional engagement is not a causal learning study. Accessibility benefits, objections to AI and the limitations of secure examinations complicate any universal prescription. [107]

Detector evidence also resists a categorical verdict. A controlled 2026 benchmark finds meaningful differences between tools; an older robustness experiment documents vulnerabilities under different conditions. Neither detector flags on unlabeled theses nor laboratory accuracy establishes an individual student's misconduct. The newer study's correction also matters: its fully

generated texts used an o3-based Deep Research system, not GPT-4o. Models, thresholds and evaluation populations must remain attached to accuracy claims. [108, 109, 110]

Common Sense Media's product assessments illustrate a further distinction: obtaining a complete homework response is not evidence of correctness, learning or observed cheating. New York City's younger-grade moratorium and Norway's age-tiered recommendations document institutional restraint, not measured benefits of those policies. The unresolved research question is which assessment designs preserve independent understanding while permitting genuinely useful assistance. [111, 112, 113, 114]

Capability, comprehension and acquired knowledge

The final Wiles study reports improved performance by consultants on unfamiliar data-science tasks, but not expert-equivalent results or improved unaided technical knowledge afterward. A separate student study finds weaker code comprehension with an editing agent than with a chatbot, despite better initial task performance. These outcomes concern different populations and constructs. Null knowledge effects are not demonstrated harm; adjustment-sensitive extension results should not be summarized as uniformly worse performance. [115, 116]

Task expansion and the interpretation of cognitive evidence

A field experiment with German IT apprentices supplies more direct evidence of AI-enabled task expansion than practitioner reports alone. It finds benefits on tasks beyond the curriculum without lower immediate comprehension. The scope matters: comprehension uses short tests, internet access remains available under instructions not to use aids, and open answers use blinded LLM grading with teacher-validated rubrics and sensitivity checks. These results do not establish durable expertise or production delivery. [117]

Recent citation-linked reviews help interpret apparently conflicting findings. Fan and colleagues propose a conceptual learning/performance framework rather than an empirically validated model; Riva and colleagues develop a narrative account of cognitive capital rather than a new longitudinal dataset. Inie's scoping review emphasizes validity and measurement, not a quality-weighted pooled effect. They add explanatory structure, not independent replications of the studies they cite. [118, 119, 120]

Cognitive relief is not the same as retained capability

Lee and colleagues' CHI 2025 survey was already listed in October; November adds substantive appraisal and its DOI, not a duplicate study. Among regular workplace users, confidence in AI and confidence in oneself have different associations with reported critical thinking. These are observational self-reports, not objective tests of longitudinal skill loss. Reduced effort can reflect effective delegation, inadequate scrutiny or different combinations across tasks. [121]

Recent interventions help distinguish these possibilities. A metacognitive-scaffolding experiment finds writing benefits despite lower perceived helpfulness; its later task removes the scaffold, not access to AI. A task-fit experiment reports efficiency gains alongside task-dependent quality, including adverse knowledge-acquisition results. Neither warrants assuming that immediate assisted performance establishes durable unaided learning. [122, 123]

An originality-incentive experiment also qualifies an inevitable-homogenization narrative: incentives can change co-creation strategies and collective diversity in a writing setting. This does

not establish that the same intervention preserves scientific diversity. A conceptual paper on artistic development and a philosophical deskilling analysis broaden the questions but supply no equivalent experimental effect size. [124, 125, 126]

Reports about changing homework assessment illustrate institutional adaptation, not by themselves learning gains or losses. Evaluate independent reasoning, delayed retention, transfer and the educational purpose of assessment alongside the convenience of generating an answer. [127]

7. Scientific discovery: prediction, search, and judgment

AI's scientific contribution spans several distinct activities: organizing literature, writing empirical software, generating hypotheses, finding algorithms, establishing formal proofs, and conducting a research process. Success in one is informative but insufficient to establish success in all.

ERA demonstrates evaluator-guided search over empirical software. Co-Scientist and Robin connect generated hypotheses or analyses with selected expert and experimental validation. The AI Scientist explores a more extended ML-research workflow, including evaluation of generated work by workshop review. Each represents a meaningful expansion of the toolchain. Their scientific credibility depends on the particular validation process, the selection of cases, and the contribution of human investigators. [128, 129, 130, 131]

Formal mathematics and algorithm search illustrate why difficult problems can be suitable for AI. AlphaProof benefits from a machine-checkable formal environment. AlphaEvolve and DiscoRL use feedback to explore algorithms or learning rules. These systems provide stronger evidence than a fluent narrative asserting a discovery, because an external procedure can reject many unsuccessful proposals. The meaning of the result still depends on the problem formulation and the domain of transfer. [132, 133, 134]

The value can also lie in connecting established methods that humans have not economically brought together. Schwartz's October account of BootLoops describes cross-disciplinary quantitative work in which experts helped distinguish a technically interesting computation from a scientifically important result. The account is a selected case study, with disclosed affiliation and ongoing validation, rather than a controlled estimate of research productivity. It nevertheless offers a useful mechanism for making previously impractical investigations possible. [135]

This distinction changes the evaluation question. It is not enough to count correct calculations or generated manuscripts. One must ask whether the result is new, whether its interpretation is sound, whether the underlying evidence supports it, and whether it matters to the field. Scientific importance is partly a collective judgment informed by replication, explanation, and the problems researchers need to resolve.

Messeri and Crockett's perspective warns of illusions of understanding and scientific monocultures. Hao and colleagues' large historical analysis supplies a related but different observational result: individual advantages associated with AI use can coexist with a narrower collective distribution of research. That study covers broader historical AI than recent LLM assistants, and should not be used as a causal estimate of chatbot-induced contraction. [20, 136]

The evidence therefore supports optimism about new computational reach and caution about equating local scientific output with the health of scientific inquiry. Systems can make difficult work more feasible while concentrating attention on questions with abundant data and

convenient evaluators. Maintaining diversity of problems and methods is part of assessing the benefit.

Mathematical capability, understanding and consent

The June *Leiden Declaration on Artificial Intelligence and Mathematics*, endorsed by the International Mathematical Union, supplies an earlier governance framework: disclose tools, retain human responsibility for correctness and authorship, credit prior work, support rigorous review, and protect research autonomy and open science. It recognizes AI's potential rather than proposing a blanket prohibition. These are community recommendations, not measured effects or evidence of unanimous agreement. [137]

Tao's 2026 ICM essay separates the arrival of capable mathematical tools from the question of what mathematics values. His October *Math 2.0* lecture recognizes powerful problem solving while proposing a broader aim: use AI to improve exposition, research networks, formal libraries and human understanding, not merely accumulate solutions. This is a substantive expert agenda, not measured proof that automated mathematics is harmful. Klowden and Tao's earlier philosophical essay supplies the human-centered framing. [138, 139, 140]

Opposition is not confined to doubtful capability. The September declaration *A Severe Misalignment of AI in Mathematics*, initiated by 25 Fields medallists, recognizes advances but contests the incentives and effects of problem-count optimization. In October, the Association for Human Mathematics rejects OpenAI's mass-release model and calls for mathematicians to discontinue collaboration with the company. These are explicit, consequential positions, not population surveys or technical disproofs. They deserve representation without being described as the unanimous opinion of mathematics. Tao's AI-assisted, personally reviewed living summary also shows that criticism of particular practices can coexist with active AI use. [141, 142, 143]

Navier-Stokes: what a proof certificate does and does not settle

On 8 September, OpenAI announced a resolution of the Navier-Stokes Millennium problem and released a written manuscript and Lean artifacts. The claim concerns smooth forced breakdown alternatives, not a demonstration of all unforced fluid behavior. We inspected the stated theorem and outline, not every lemma or the running proof kernels. Clay's 11 September statement welcomes the apparent resolution while retaining its deliberate evaluation process; it is not a prize award. [144, 145, 146, 147]

Bastounis, Circelli and Hansen's 6 October preprint reports differences between the written argument and Lean lemmas, including derivative requirements. Its central warning is that a checked formal argument need not faithfully reproduce an accompanying natural-language argument. The authors explicitly do not adjudicate the correctness of the written proof. The theoretical worst-case result does not make every practical translation impossible. Nor does a discrepancy alone establish that the formal theorem is false. We retain the developer claim and the critique as distinct validation questions that this review does not resolve, rather than independently certifying or rejecting the claimed solution. [148]

Hales provides a complementary technical account: kernel validity, statement fidelity, independent checking and readable exposition are different assurance layers. Lean's maintainer documents a positive AI contribution to finding and repairing soundness bugs, while treating adversarial proof artifacts as a genuine risk. Together these accounts show why formal mathematics is promising without being verification-free. [149, 150]

Scientific feedback, selection and physical execution

A randomized feedback-delivery experiment reports more manuscript revisions, not independently validated improvements in science. A retrospective grant comparison raises a different concern: model selection diverges from human decisions for proposals humans rate as novel, under a design that does not provide the same explicit rubric to the model. Neither establishes the universal value of AI scientific judgment. [151, 152]

AutoDNA's reported physical laboratory demonstrations broaden the evidence beyond digital artifacts. They concern a co-designed system with ongoing human maintenance, not autonomous general science; the public abstract was inspected, not complete subscription methods. [153]

Scientific reuse, understanding and institutional choice

Scientific help includes making existing methods easier to inspect and reuse. Paper2Agent evaluates executable research resources on tutorial-derived tasks; success there is not independent confirmation of the original science. A large AI-assisted analysis of research documentation similarly estimates a reproducibility proxy rather than reporting thousands of successful reruns. These intermediate contributions deserve recognition without conflating reuse with validation or autonomous discovery. [154, 155]

The later October mathematical announcements and reporting add a different evidentiary category. Claims about Yang-Mills and ETH require precise statements, accessible artifacts and independent scrutiny. Reporting about an existence result is not certification of the full mass-gap problem; a reported ETH refutation is not P versus NP or a demonstrated practical cryptographic break. This review has not independently checked the proofs. Developer announcements and news coverage of the same event are not separate replications. [156, 157, 158]

Strogatz's essay distinguishes obtaining results from understanding, learning and participation. Calegari's guest post on Tao's blog proposes institutional changes in response to mathematical capabilities; the guest author should not be mistaken for the host. Altman's reported rationale for continued releases supplies the developer's perspective, not independent validation of discoveries or their cost-effectiveness. These positions can inform choices about stewardship without becoming a popularity vote on technical correctness. [159, 160, 161]

The knowledge ecosystem also matters. Wikimedia's proposed court intervention differentiates training from source-substituting retrieval, with qualifications for licensed and fact-dependent uses. This is advocacy, not a settled legal rule or measured productivity result. The broader evaluation question is whether assistance preserves the people and institutions needed to produce future knowledge, not merely whether it makes current information easier to retrieve. [162, 163]

8. Can AI improve the next generation of AI?

AI already contributes to components of AI development, but several layers should be distinguished. Optimizing a kernel, improving training data, proposing a learning rule, changing an agent's tools, and building a new foundation model are different activities. They require different evidence of transfer and autonomy.

Darwin Godel Machine and more recent harness-evolution work demonstrate a route through

changes to agent software. A frozen model can power an improving system when the surrounding prompts, tools, memory, and control flow change. Tool-architecture experiments reinforce this point: agent behavior and costs depend on the interface through which capabilities are exposed, not only on the underlying model. [164, 165, 166]

InnovationEval asks a harder question about independent algorithmic research. The October report evaluates whether agents can develop a post-training method comparable to a withheld human innovation. The tested configurations make limited progress, and the report identifies misleading selection of favorable runs. Its interpretation is bounded by one task, few runs, scope judgments, and successor-model contamination. [26]

ResearchGym also distinguishes occasional strong results from reliable long research workflows. A single success can establish possibility without making the process dependable. Conversely, a failed benchmark run cannot establish that no useful AI-assisted research exists. [167]

Automated alignment research offers a positive contrast under another protocol: agents improve metrics for defined unwanted behaviors, with selected held-out and transfer evaluations. The human comparison does not give participants the same iterative opportunities, and benchmark improvements are narrower than overall production safety. The two studies therefore sharpen the conditions for progress rather than settling an unrestricted question about autonomous intelligence. [168]

Anthropic's internal R&D measurements add a process view, distinguishing assistance, collaboration, leadership, and autonomy. Their task basket is model-rated and their numbers require external checking. They should not be converted into a percentage increase in discovery speed. [169]

Our assessment is that AI-supported AI development is already a substantive engineering reality. The evidence for reliable end-to-end innovation is less settled. The appropriate questions concern what improved, how it was tested, how far it transfers, and which parts of goal setting and evaluation remain outside the automated process.

Generation and self-evaluation remain different capabilities

Glazner's engineering essay locates value in tacit experience and responsibility; this is a hypothesis about human contribution, not evidence of an immutable boundary. Mollick's selected cross-domain demonstrations show both compelling generation and substantial steering needs. His AGI label is a restricted interpretation, while the research demonstration is explicitly not fully human-validated. Neither a polished artifact nor agent review alone establishes correctness, durable value or representative expert-level performance. [170, 171]

AI-assisted AI research: acceleration is not autonomy

METR's high-access Anthropic assessment estimates 1.9 times the earlier growth rate of an internal capability index, with a reported 95% interval of 1.5-2.8. Its attribution to AI is a separate judgment with substantial uncertainty and largely redacted evidence. The estimate is neither a randomized researcher-productivity effect nor proof of autonomous recursive improvement; METR's earlier methodological proposal separates these questions. Cotra's compounding scenario is interpretation of the same assessment, not independent evidence or METR's forecast. [172, 173, 174]

This does not straightforwardly contradict InnovationEval's weak end-to-end autonomous re-

sults: assistance inside an expert organization and autonomous performance on a selected benchmark measure different capabilities. Both deserve attention. [26]

Scientific reporting needs claim-to-experiment provenance

Into the Unknown tests agents on expert-specified research follow-ups and reports no submission satisfying every criterion. Its strict operational rubrics and automated judges are part of the evaluation design, not a universal standard for all scientific assistance. A subsequent reanalysis of the same artifacts distinguishes unsupported numbers from real values assigned to the wrong experiment. Judge choice changes absolute pass rates, and automatic non-certification does not establish falsehood. These are complementary analyses of shared outputs, not separate replications. [175, 176]

The recent journal publication of Luo and colleagues' workflow-pitfall study also makes process visibility salient; its earlier preprint is the same study family. Fujimura's small demonstration offers a constructive counterpart through consecutive agent-selected research runs and re-computation, while disclosing post-result amendments, AI-only ratings and the lack of independent human validation. Re-running code, demonstrating novelty and producing scientifically sound conclusions remain different achievements. [177, 178]

A new scientific-agent methods primer provides a broader workflow map. Only its public abstract and metadata were inspected here, so it informs orientation rather than detailed claims about effectiveness. [179]

9. Business value, cost, and organizational change

KPMG's quarterly executive surveys illustrate the gap between reported benefits and defensible economics. The June Q2 announcement reports established ROI for 7% of respondents. The September Q3 report, by Priya Emmanuel, Benedikt Höck and Simon Benson, surveys 2,131 senior leaders: 89% report measurable AI value, but the headline share consistently assessing value against cost is only 12%; 10% describe their organization as having established ROI. These are different measures, not conflicting estimates of the same outcome. [180, 181]

The Q3 survey covers large organizations, not the typical small firm, and its value/cost questions exclude early-stage respondents. Group differences associated with controls or accountability are observational, not proof that those practices caused returns; maturity itself includes reported ROI. Comparisons across quarterly waves are not verified within-firm changes. The useful signal is the need to connect benefits with costs, not an independently audited return on AI investment. [181]

The business question includes the total cost of obtaining an accepted result: user time, model inference, compute experiments, integration, verification, rework, and later operation. Counting tokens or licenses can describe input consumption without establishing benefit. Reported spending restrictions at Uber are useful as a case about value attribution, but not an audited estimate of whether AI paid for itself. [182]

Sustainability broadens that accounting. Iqbal and colleagues combine a literature review with practitioner interviews and identify efficiency benefits alongside energy-monitoring gaps, dependency, skill and defect concerns. Their qualitative design does not determine a net emissions balance. The implication is to measure the resources and risks shifted elsewhere, rather than

assume that faster coding reduces every cost. [183]

Benefits may reach an organization through changed demand or an expanded product range, rather than the same work being performed with fewer employees. Retail field experiments provide evidence closer to commercial outcomes, finding heterogeneous sales effects across integrations. Organization-capital research proposes another channel: AI investment can build durable firm-specific knowledge and processes that improve production. The latter work studies broader AI investment and relies on firm-level proxies, so it does not establish the return from purchasing a chatbot subscription. [184, 185]

It is useful to distinguish **efficiency value**, in which existing work becomes cheaper; **quality value**, in which decisions or artifacts improve; and **option value**, in which a previously uneconomic project becomes worth attempting. These are analytical categories used in this review. They are not interchangeable accounting measures and cannot all be inferred from a faster task completion time.

Option value is particularly important for difficult intellectual work. The economically relevant comparator can be a simplified analysis, a small experiment, or no project at all. An AI-assisted system that makes a more ambitious study feasible may create substantial value even if every result still requires expensive expert validation. A narrow task-speed experiment would miss that change in the feasible project set.

Organizations should therefore specify the intended outcome before comparing tools. For an existing workflow, the test may be cost per accepted result and defect-adjusted throughput. For an exploratory project, it may be the number of credible hypotheses tested within a budget or the probability of obtaining a decision-relevant result. Evaluations should preserve failures and abandoned attempts, because a collection of only successful examples cannot estimate the return to experimentation.

Public discussion adds context, not additional controlled effects. The Pragmatic Engineer's reader survey describes heterogeneous experience and cost concerns; TechTarget connects verification and comprehension risks; Gates argues for managing disruption and distributing benefits. Market commentary describes investor expectations, which are not realized productivity. Ars Technica's bottleneck account covers the Chen-Stratton study already analyzed here and must not be counted as another independent result. [186, 187, 188, 189, 190]

Organizational results require task-specific designs

A revised randomized voice-interview study finds higher hiring-related outcomes when AI collects information and humans retain decisions. Its entry-level recruitment setting does not establish fairness or effects in expert hiring. A new small-firm e-commerce study reports sales benefits under an observational identification design; only its primary indexed abstract was accessible here, so detailed causal appraisal remains pending. These are more direct organizational questions than counting generated artifacts, but their denominators still differ. [191, 192]

Agent-cost research also shows why implementation matters: guidance can reduce benchmark inference spending, while additional retrieval infrastructure can increase it. Neither maximum token savings nor benchmark success is total organizational return. [193]

Commercial success is a separate denominator

Raso's optimistic valuation argument is useful as a counterpoint, not a settled financial assessment. The linked company filing distinguishes revenue, compute expense, operating loss and noncash valuation changes; exceeding one expense category does not establish profitability. Infrastructure also entails continuing commitments and replacement costs. Provider growth, investor returns, customer value and worker welfare must not be treated as interchangeable evidence that AI helps. [194, 195]

Complementarity is a hypothesis, not a guarantee of payoff

The Productivity Institute's commissioned synthesis connects task, job, organization and economy, emphasizing complementary investment and potentially delayed returns. Its eight expert interviews and literature synthesis are not a new causal trial. The productivity J-curve is a possible mechanism; it does not show that every disappointing investment will eventually pay. Sponsor interests, earlier study versions and overlapping evidence must remain visible. [196, 197]

McKinsey's 2026 survey separates perceived individual productivity from reported enterprise financial effects. Neither an absence of reported EBIT impact nor a dispute about the widely repeated 95% failure statistic establishes a universal pilot-failure rate. Macroeconomic commentary, executive surveys and experimental task results use different denominators and time horizons. [198, 199, 200]

Amazon's engineering memo favors outcomes, warns against token and code-volume targets, yet retains adoption metrics as diagnostics. Its recognition of legitimate non-benefit findings creates an important governance test: can workers report adverse results without penalty? The memo is evidence of stated expectations, not a causal productivity estimate. Cisco's reported modernization proposal belongs in a prospective case register, not the pool of completed successes. [201, 202]

Unit prices are also incomplete. Provider pricing and practitioner analyses help define cost boundaries; modeled inference efficiency does not settle expenditure per accepted result. Cloud memory-allocation arguments and manufacturer forecasts add questions about access and control, but do not establish a universally superior allocation of resources. Count retries, checking, hidden agent calls and downstream costs, then ask who receives the benefit. [203, 204, 205, 206, 207]

10. Work, employment, and the future supply of expertise

Productivity evidence and employment evidence answer different questions. Labor demand depends on output demand, organizational expansion, substitution between tasks, wages, regulation, and the speed at which practices change. A real task benefit need not immediately change headcount or earnings.

Humlum and Vestergaard link Danish adoption surveys to administrative records and find substantial task reorganization alongside limited early average earnings and hours effects. U.S. payroll and vacancy research raises concerns about entry-level pathways in exposed occupations. These differences call for comparison of institutions, observation periods, and units of analysis. They do not justify choosing one country as the universal outcome. [208, 209, 210]

Ramp and Revelio provide a counterpoint at the firm level: high-intensity adopters show stronger employment growth, including entry-level hiring. The adopters already differ in size, growth, technical intensity, and financing. That result should temper universal displacement claims without becoming a claim that higher AI spending causes hiring everywhere. [211]

There is also a developmental pathway between education and expert work. If organizations delegate or eliminate the tasks through which novices traditionally acquire knowledge, they may need to create new forms of supervised practice. This is a plausible institutional risk, not a demonstrated inevitable shortage. Herrengt's essays articulate a stronger hypothesis about the erosion of intermediate engineering roles; they are useful viewpoints whose labor-market predictions require empirical testing. [212, 213]

Educational choices provide an earlier signal of changing expectations. U.S. enrollment data report declines in traditional Computer and Information Science alongside growth in data-oriented programs. UK official data for a different period show growth in Computing and AI. These contrasts caution against treating changes in course selection as direct evidence of AI substitution. They may reflect employment conditions, international mobility, program naming, or perceived returns. [214, 215]

The key long-run question is whether AI adoption leaves enough people able to understand and improve the systems on which institutions depend. That question includes access to training, meaningful work for entrants, concentration of expertise, and the distribution of benefits across workers and firms. Current evidence is more informative about early transitions than settled long-run outcomes.

Welfare is not a residual productivity metric

Recent Ghanaian adult-learning research records both perceived benefits and difficulties in a setting rarely covered by engineering trials. Its cross-sectional measures and purposively selected narratives do not establish causal health or learning effects. Gallup's November-updated synthesis likewise concerns perceived productivity, not observed output. These perspectives deserve inclusion without displacing telemetry, experiments or the voices of people who reject assistance. [216, 217]

Longer workdays are not necessarily greater output

ActivTrak's October release compares 9,140 workers and reports AI-user workday spans up to 90 minutes longer than non-users'. This is an association, not evidence that AI caused additional labor or reduced productivity. Roles and workloads may differ; span is not active working time, and focus comparisons use different groups. Its linked annual report covers earlier periods and separate cohorts, so it is not interchangeable with this analysis. Neither telemetry nor subjective enthusiasm alone establishes welfare or accepted output. [218, 219]

Taken alongside qualitative work-intensification research, this is a reason to measure whether assistance buys time, expands chosen work or increases involuntary demands, without presuming that those outcomes coincide. [220]

Adoption, recruitment and the distribution of work

The October revision of Firm Data on AI expands its executive sample to nearly 9,000 respondents in seven countries. Retrospective assessments, expectations and offsetting employment

changes across firms are not audited causal estimates. Task-adoption research further distinguishes actual reported use from occupational exposure and provider chat logs. Public abstracts and metadata, not complete papers, underpin these two intake appraisals. [221, 222]

Accenture's investor presentation reports recruitment and onboarding gains; this is interested-party testimony without a disclosed comparison design or sufficiently specified quality measure. A human final hiring decision does not establish fair upstream screening, and time to billable work is not retained unaided skill. A historical recruitment critique helps formulate fairness questions but does not audit Accenture's current systems. [223, 224]

Employment announcements need equally careful boundaries. Microsoft's voluntary program and the Dutch government's multi-year FTE target are not causal estimates of AI displacement. The Bank of America report combines remediation and compliance expense; contractor additions cannot be treated as net employment gains or model-only costs. Reuters' account of Indian IT contracts adds distributional questions about client bargaining and provider revenue, not a universal productivity rate. [225, 226, 227, 228]

More feasible work need not free workers' time. An ethnographic account describes intensification and boundary erosion, while a staged Copilot rollout studies changing collaboration networks. Communication time is not itself value, just as workload growth is not automatically worker welfare. A global ILO analysis adds the constraint of unequal digital access: potential task exposure is not realized augmentation or observed displacement. [220, 229, 230]

11. An integrated assessment: where the benefit survives

We propose a practical evaluation pathway: **generation, verification, integration, use, and value**. It is a conceptual synthesis, not a fitted causal model. A gain can occur at each stage, and a constraint at one stage can absorb improvements at another. The pathway also has feedback: real-world outcomes can alter the next task, specification, or research question.

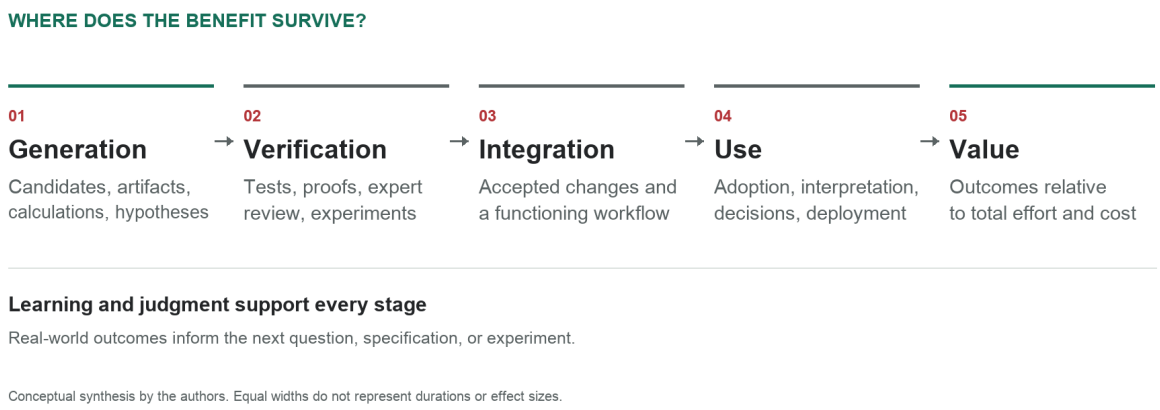


Figure 1. Evaluation pathway for AI-assisted intellectual work. The stages are conceptual; their equal visual widths do not represent equal durations or estimated effects. Source: authors' synthesis.

The framework suggests five questions for a particular deployment:

1. What becomes possible or faster, and relative to which realistic alternative?
2. What evidence makes the result acceptable, and what does verification cost?

3. Can the surrounding workflow integrate the additional output?
4. Is the accepted output used, and does it improve a consequential outcome?
5. What happens to the knowledge and judgment required for the next round of work?

A deployment can succeed without full autonomy. It can also be valuable without producing a large employment effect. Conversely, broad adoption, strong benchmark scores, and enthusiastic user reports remain insufficient to establish the complete value pathway.

The evidence is strongest for specific task and workflow benefits, increasingly informative about downstream engineering constraints, and still heterogeneous on learning and labor effects. Scientific benefits range from well-established specialized predictions to emerging agent workflows with very different degrees of validation. The largest opportunity may include work previously regarded as too complex or expensive to pursue, but its value must still be established through credible results and relevant outcomes.

Our answer to "Does AI Help?" is therefore **yes in demonstrated settings, no in some tested settings, and unresolved in many others**. AI already helps people produce, search, calculate and explore; it has also measurably slowed expert work, weakened some learning or comprehension outcomes, and imposed costs on reviewers and institutions. Gains and losses can coexist in the same intervention. This collection does not establish whether positive or negative net effects predominate across intellectual work: the outcomes and populations are too different, and selection is not exhaustive. The next research priority is measuring those relationships over time, rather than extrapolating one task's percentage to an entire institution.

Open-source benefits depend on provenance and reciprocity

The Asahi/Omarchy M controversy extends the question from faster development to who receives credit and sustains the underlying work. Omarchy explicitly frames its Apple Silicon effort as finishing what Asahi started. Asahi subsequently announces stronger copyleft, alleging unattributed reuse and LLM-assisted concealment without naming an accused project. DHH defends licensed reuse and says the AI contribution ban prevents upstreaming. The public connection is substantial, but specific misconduct is not established by these statements alone. [231, 232, 233]

Funding gives the reciprocity dispute material context: reported backing includes multiyear pledges and AI tokens, not only received cash. Repository evidence also complicates a simple narrative: Omarchy acknowledges its Asahi base, while an inspected installer pins an older MIT-licensed m1n1 revision despite that component's later GPL transition. This is selected provenance evidence, not a compliance certification. [234, 235, 236]

License permission, retained notices, community credit, contribution acceptance and clean-room provenance are distinct obligations or expectations. Neither an LLM rewrite nor a functioning product establishes independent origin; copyleft is not a general ban on commercial forks or AI. Our interpretation is that assistance can benefit downstream users while redistributing credit and maintenance costs. This case exposes that possibility, not its measured net effect. [237, 238, 53]

Adoption and expertise have institutional denominators

Lane's October ECB keynote, citing the revised Firm Data on AI, brings European firm and worker surveys into this discussion. Its usage intensity, perceived savings and investment measures concern different populations and periods; macroeconomic scenarios are not realized effects. In particular, respondents' total weekly savings should not be added across tasks as if they were independent increments. [239]

Ranade's guest essay on Tao's blog identifies intellectual training and participation as a public-good concern. This is an institutional argument, not an employment estimate. The question of who finances the development of future expertise remains important even without accepting categorical claims that AI augmentation cannot support human understanding. That latter question requires empirical assessment of particular learning designs. [240]

What the November additions do and do not establish

The evidence increasingly supports evaluating capability expansion, accepted delivery, human development, economic value and institutional sustainability separately. It does not establish an inevitable positive or negative trajectory. Stronger tools can remove one constraint while revealing another; useful discovery can create costs; incentives and access can alter who benefits.

This preview includes all staged records in the evidence map, while citing selected sources in the synthesis. It distinguishes full-text, selected-section, abstract-only and preview-only inspection. Older access observations are not silently renewed. Monthly browser checks, outstanding full-text follow-ups and final editorial approval remain requirements before a formal November release.

12. Review method, authorship, and release policy

Scope and selection

This is a living critical narrative review with an accompanying evidence map. It concentrates on 2023-2026 research while retaining selected earlier foundations. The focus is intellectual work: study, programming, engineering, scientific investigation, and related organizational outcomes. Specialized AI is included where it clarifies a mechanism or provides a credible scientific comparator; findings about specialized models are not automatically attributed to general-purpose LLMs.

The initial collection was supplied and developed over time, then supplemented through primary-source searches of arXiv, journal and publisher sites, research organizations, and selected expert analyses. Searches were organized around productivity experiments, agentic engineering, learning and transfer, AI-supported science, business outcomes, and labor transitions. This release does not claim PRISMA compliance, exhaustive coverage, a pre-registered search strategy, or a pooled meta-analytic effect. Coverage is weighted toward English-language sources and the settings in which current evidence is most visible.

Current-release access limitations

Four contextual references could not be reverified for November because their websites deny access or their archive fails: the Daily Californian account of homework assessment and three Linux mailing-list messages on Assisted-by attribution. They are retained with their earlier dated reading history, not presented as newly confirmed sources or independent research evidence. The human coauthor approved keeping these records explicitly unverified and nonblocking for this release; they will be revisited in the next monthly review. [127, 92, 93, 94]

Verification and appraisal

The targeted pre-publication search for companion reviews is documented in the search and reading log. It adds complementary syntheses without claiming exhaustive coverage or counting their constituent studies as independent new evidence.

Every database record distinguishes **content-checked**, **identity-checked**, and **candidate** status. Content checking records the inspected scope: in some cases a full paper or methods section; in others an abstract or an official summary. Identity checking establishes bibliographic correspondence, not the truth of findings. Candidate records preserve incomplete access or unresolved identification without treating the source as verified evidence. A working link alone does not satisfy result verification.

Core annotations contain an original short summary and limitations. Supporting records have bibliographic and topical descriptions but do not assert unappraised findings. Method, publication status, and relevance are distinct fields. An arXiv DOI identifies a preprint and does not establish peer review. Multiple pages or versions of a work are retained as linked reference records where appropriate, but are not counted as independent replications. The exclusions log records rejected inherited links.

We prefer primary reports for empirical claims. Vendor research and expert essays remain useful when their evidentiary role and interests are explicit. Paywalls and access failures are reported rather than bypassed. Numeric claims are tied to source versions, populations, outcomes, and denominators. Negative or null results receive the same attention as positive findings; statistical insignificance is not automatically evidence of equivalence.

A separate balance self-review and source-admission log documents evidentiary balance. It distinguishes demonstrated harm, imprecise or null effects, mixed outcomes, forecasts, and community refusal. Equal evidentiary standards do not require equal numbers of positive and negative paragraphs. We do not infer an overall verdict by counting favorable sources, treating duplicate coverage as replication, or treating community opposition as a measured causal effect. Both AI authorship and the human author's business affiliation warrant scrutiny, not a presumption that this review is neutral by construction.

For selected sources that automated CI cannot reliably retrieve or identify, we use a separate local Chrome verification channel. The browser ledger records the AI-assisted reviewer, observed metadata, exact reading scope and access limits. It is refreshed for each monthly release; previous observations are preserved. A browser confirmation can establish an otherwise unresolved citation, but cannot override an identifier conflict, certify a scientific claim or imply access to subscription full text.

Chang and Huang's journal paper offers survey-based evidence about AI reliance and perceived development performance; only its public preview and snippets were inspected. CNBC's em-

ployment account is secondary reporting, not an independent causal study. These reading and methodological limits constrain their evidentiary weight. [241, 242]

Document inspection includes supplied PDFs matched to their online source versions. Filenames, checksums and exact reading scopes are recorded without redistributing third-party PDFs. The document inspection log documents title/version differences, assistance-design boundaries and reporting questions for the chess-training, AI Skill Threat and agent-study materials. The Financial Times source is an unappraised candidate because full-text access was unavailable. This access limit is not evidence for or against the article's claims.

Greg Wilson's methodological discussion is a useful companion for readers evaluating common overclaims in AI-assisted coding. The more general lesson is to inspect study design and outcome validity before accepting a headline about productivity. [243]

The CoSLR study supplies a directly relevant caution for AI-assisted evidence synthesis: mandatory checkpoints did not ensure that participants intended to verify generated reports. Its short, survey-based evaluation does not measure actual review accuracy, but motivates explicit citation checks and substantive editorial scrutiny here. Our verification scripts establish bibliographic correspondence, not whether every interpretation is sound. [244]

Authorship and responsibility

GPT 6.1 is credited as the lead AI author, responsible for organizing the literature, appraising the inspected evidence, developing the synthesis, drafting the article, and preparing the publication materials. **Witold Bołt is credited as co-author**, responsible for initiating the topic, providing initial links and viewpoints, framing the question, directing the conceptual scope, and initiating this review in its present form.

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